

LearnPath: Auditable Adaptive Diagnosis and Coverage-Aware Assessment Scheduling

Dr. K Anand
Rajalakshmi Engineering College
Chennai, India
anand@rajalakshmi.edu.in

Hariharan N S R
Rajalakshmi Engineering College
Chennai, India
230701392@rajalakshmi.edu.in

Udhay Kumar V
Rajalakshmi Engineering College
Chennai, India
230701367@rajalakshmi.edu.in

Varun S P
Rajalakshmi Engineering College
Chennai, India
230701373@rajalakshmi.edu.in

Yokeshwaran K
Rajalakshmi Engineering College
Chennai, India
230701389@rajalakshmi.edu.in

Abstract—Adaptive learning systems are only as trustworthy as the learner estimates behind them, and those estimates should show how much evidence supports them. This paper describes LearnPath, an implemented diagnostic architecture that keeps mastery, confidence, assessment coverage, and prerequisite readiness as separate quantities. Self-reports decide which questions get asked but never count as mastery evidence, and short bounded sessions feed a persistent course-wide assessment program. We also test whether preference-based question selection keeps candidate items available and improves evidence coverage, using controlled response replay: six policy configurations on 4,096 paired synthetic response matrices, giving 24,576 session runs. The deployed selector ends sessions prematurely in 47.02% of these cases. An eligibility-first variant removes the failure without touching the scoring or confidence rules, yet its assessed-skill coverage (33.81%) still trails a stratified twelve-question baseline (37.77%). The results expose a scheduling defect and a bounded fix; they do not show adaptive superiority or diagnostic accuracy, and we make no claim about human learning outcomes. Finite invariant checks and planner traversal complement the replay.

Index Terms—Adaptive diagnosis, assessment coverage, evidence provenance, knowledge graphs, learner modeling, personalized learning.

I. INTRODUCTION

A personalized learning system has to answer two questions that are easy to run together: what should this learner study next, and what can the system honestly claim to know about them? Recommending an advanced topic is risky when its foundations lack evidence, yet repeatedly serving introductory material the learner already knows is just as unhelpful. The hard part is building a usable picture of the learner from evidence that is partial, mixed, and sometimes contradictory.

Large curricula make this harder. A fixed entrance quiz can sample only a small share of a course with many skills, and treating the unsampled skills as failures mistakes missing evidence for evidence of a gap. The opposite error is just as easy to make: propagating a correct advanced answer back through a prerequisite graph invents foundations nobody observed. Learners may recall a sophisticated procedure yet have lost a basic operation, so a diagnostic model needs room

for such cases instead of forcing every learner onto one line from beginner to expert.

LearnPath takes an evidence-bounded approach. It gives each learner a provisional starting region, keeps unresolved skills on record, and continues assessment over bounded sessions and later learning activity. Self-reported familiarity shapes which evidence the system goes looking for, but only scored responses are recorded as evidence. Skills have a shared identity so evidence can be reused across courses, while the current course decides which skills and prerequisites matter.

Estimation and item choice are well studied in knowledge tracing, adaptive testing, and constrained selection [1]–[6]. The problem we address is a narrower engineering one that surfaced in our own system: applying a preference filter before checking eligibility can empty the candidate pool. A preferred skill that has hit its observation cap can hide another skill’s usable item, and the selector then reports an exhaustion that does not exist. We isolate this ordering defect and do not present eligibility or uncertainty as new research topics.

We ask three questions. RQ1: does the selector return an item whenever at least one unasked item is still eligible? RQ2: how does assessed coverage compare with stratified selection at fixed and at matched question counts? RQ3: how do coverage and failure behavior change with bank depth, response propensity, self-report, and graph metadata? None of them tests diagnostic validity.

Our contributions are: (1) an auditable architecture that keeps mastery, confidence, coverage, provenance, retention, and prerequisite readiness apart; (2) the identification of the preference-before-eligibility failure, with a correction that preserves availability; and (3) a paired synthetic replay of deployed, corrected, stratified, and ablated configurations, reported together with the cases where the correction does not help. The correction has been evaluated offline only and has not replaced the deployed policy.

II. RELATED WORK AND POSITIONING

Knowledge tracing infers skill acquisition from observed performance [1], and Deep Knowledge Tracing applies recurrent networks to learning histories [2]. LearnPath uses neither; its update rules are explicit and not fitted to an independent cohort. Evidence-centered assessment ties the assessment argument to its evidence and operational processes [7]; LearnPath keeps those records inspectable, though traceability alone does not show that its items measure the intended constructs.

Computerized adaptive testing (CAT) offers information-based selection [3], severely constrained selection [4], and constrained full-test assembly [5]. Shadow-test methods handle exposure control through item-ineligibility constraints [6], and Han separates content balancing, the selection criterion, and exposure control [8]. LearnPath does not claim to have introduced any of this. Our contribution is the narrower property that a preference filter must not remove every usable alternative while local eligibility is nonempty. We do not solve globally feasible test assembly or formal exposure control, and our information proxy uses author-supplied priors rather than Fisher information; the exposure penalty and fixed weights are engineering rules, not a calibrated CAT model.

Knowledge-space theory describes the feasible knowledge states of a domain [9]. LearnPath instead treats authored prerequisite edges as learning constraints and assessment priorities, and a correct advanced answer never establishes the prerequisites beneath it. Ngo *et al.* recommend learning paths by combining graph embedding, sequence mining, and collaborative filtering [10]; their task differs from ours, since we gather the evidence a path planner consumes, and we do not reproduce their model as a baseline. Uncertainty-aware knowledge tracing uses stochastic representations [11], and domain-informed models can propagate mastery and uncertainty through subjective logic [12]. We claim neither predictive superiority nor novelty in uncertainty handling, and an explicit confidence formula is not calibration, which needs empirical agreement with observed outcomes [13]. Trainable spaced repetition [14] targets recall over time, whereas our cold-start replay only reads retention metadata. We evaluate item availability and internal evidence coverage, not prediction accuracy or learning effectiveness.

III. PROBLEM FORMULATION AND ARCHITECTURE

A. Curriculum and Learner State

We model the curriculum as a graph $G = (V, E)$. Each vertex is a globally identified skill, and a directed required edge from a to b means that a supports access to learning b . A course picks an ordered subset of V , groups it into modules, and sets target mastery thresholds. Several courses can reference the same skill identifier, but authors must confirm that the construct and expected level truly match before evidence is reused.

$$x_{u,s} = (m_{u,s}, c_{u,s}, z_{u,s}, a_{u,s}, r_{u,s}, H_{u,s}). \quad (1)$$

In (1), m is mastery, c confidence, z a diagnostic classification, a assessment coverage, r retention state, and H the evidence history including provenance. Prerequisite readiness is derived separately from the required edges and the learner evidence that supports them. These quantities are not interchangeable: mastery and confidence are bounded summaries rather than calibrated probabilities, a null mastery means no direct knowledge evidence, and the internal first-update prior of 0.5 does not represent measured 50% knowledge. The aim of a session is to reduce consequential uncertainty within a budget, not to maximize skills labeled mastered: find a defensible starting point, respect required prerequisites, and keep unresolved questions for later evidence.

B. End-to-End Processing

Fig. 1 shows the implemented division of responsibilities. A React front end collects goals and self-portrait responses and presents bounded assessments. A TypeScript/Express service manages program state, selection, scoring, and learner-model updates, with curriculum mappings, item metadata, attempts, snapshots, and evidence in PostgreSQL. A personalized-path service reads the resulting global skill state; the separate Python recommender in the wider application is not the diagnostic estimator studied here.

A learner picks a course and reports familiarity, and the assessment program schedules a small skill scope. Within a session the diagnostic service selects an item, records the response, recomputes priorities, and applies its stopping policy. On submission, answers are scored against the attempt snapshot, evidence is aggregated per skill, global mastery and confidence are updated, and affected paths are refreshed.

C. Self-Portrait and Evidence Provenance

In the self-portrait, learners rate familiarity from never learned to very comfortable and add confidence and experience context. Reported knowledge raises verification priority and reported unfamiliarity favors an anchor probe; neither grants mastery, removes a prerequisite, or completes coverage. Whether reported familiarity was borne out by performance can also be recorded, which explains why a learner was asked to verify a skill they called familiar.

Each active question carries a skill mapping, diagnostic role, cognitive level, difficulty, response format (single choice, multiple selection, or numeric), discrimination and guessing priors, a construct identifier, and a content version. Attempt snapshots preserve the question and scoring metadata as presented, submitted answers are immutable, and each diagnostic result keeps before-and-after learner summaries, evidence strength, and a decision explanation. Historical backfills and editable self-report records still need care when rebuilding a fully version-consistent study dataset.

IV. EVIDENCE-BOUNDED DIAGNOSTIC METHOD

A. Progressive Coverage of Large Courses

An outer assessment program keeps a persistent backlog over the whole course. A session covers at most four skills and

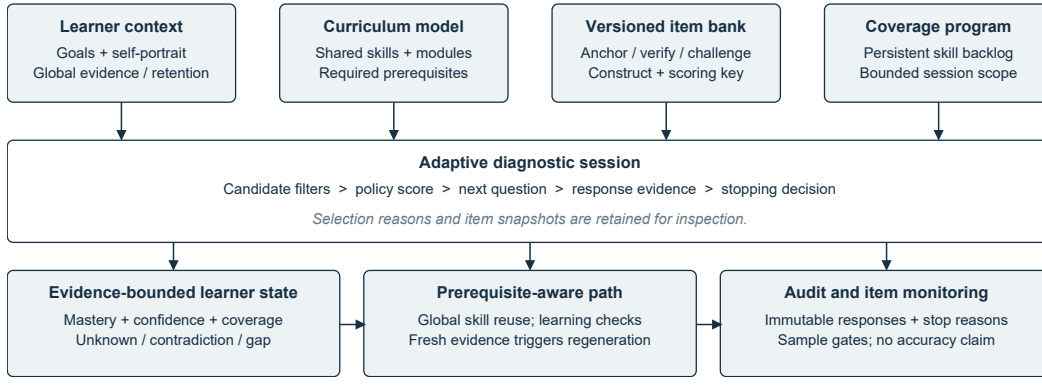


Fig. 1. LearnPath diagnostic architecture. Self-report and curriculum metadata guide collection; scored responses, not self-reports, supply diagnostic mastery evidence. Path and monitoring services consume the recorded result.

respects the diagnostic question limit. Comprehensive mode groups work by module, whereas quick placement samples curriculum regions far apart instead of stopping at the first weak foundation. Skills with unresolved evidence or retention risk stay eligible for later sessions, and since there is no fixed session limit, the per-session question budget is not a budget for the whole course. Coverage counts distinct question identifiers and evidence diversity, so repeating an item adds nothing. The usual positive route needs three distinct items, application evidence, two difficulty bands, and confidence of at least 0.50. Consistent diagnostic-only evidence has a separate route at 0.45 that does not certify mastery, and previously verified evidence has its own. A confirmed GAP or FORGOTTEN establishes coverage from two distinct items. Running out of fresh items does not by itself confer coverage. The architecture is subject-agnostic at the level of skill identifiers, but the current item formats suit quantitative reasoning best; essays and open-ended judgement are not assessed automatically.

B. Adaptive Item Selection

Within a session the policy moves through coverage, confirmation, and verification stages. Coverage avoids resampling an observed skill, confirmation seeks another independent observation, and verification favors challenge or application evidence and revisits mixed responses. Concretely, the deployed selector removes items already asked, prefers unobserved skills during coverage and contradictory skills during verification, prefers the stage-specific role within each skill (with a fallback), scores the remainder with (2), discards non-finite scores, and ranks by score, course order, and item ID, returning null for an empty pool. Curriculum role and cognitive level are separate fields, because a challenge label alone does not prove that an item measures application; the policy favors application or analysis evidence when it is missing.

$$S(q) = 100 [\mathbf{w}^T \mathbf{f}(q) + b(q) - e(q) - \tau(q)]. \quad (2)$$

The feature vector in (2) has nine components: course relevance, decision impact, evidence diversity, gateway influence, an information proxy, misconception value, retention risk,

uncertainty, and self-report priority, with weights 0.10, 0.15, 0.07, 0.15, 0.25, 0.05, 0.08, 0.15, and 0.05 in that order. The weights sum to 1.05, so S is a relative policy score and not a probability. The verification bonus b is 0.18 for mixed evidence, the exposure penalty e is capped at 0.16, and switching skills costs $\tau = 0.035$. These values predate the replay and were not fitted to its outcomes, but the repository does not record why they were chosen, so they have no empirical justification; calibration and targeted ablation remain future work.

$$I(q) = \text{clip}_{[0,1]} \left(\frac{d(q)[1 - g(q)]}{1.8} \right). \quad (3)$$

Here $d(q)$ and $g(q)$ are the item’s discrimination and guessing priors, and (3) is our information proxy. Gateway influence is the clipped dependent count divided by four. Uncertainty rises when mastery is absent, confidence is low, only one response exists, or responses disagree. Evidence diversity rewards a new cognitive level, missing application evidence, and fit to the desired difficulty. These design choices need ablation and calibration before any claim of optimal selection. A skill is limited to three observations, or four when evidence is mixed or application evidence is missing. Prior knowledge counts as recognized only if mastery reaches the course target, confidence is at least 0.70, and no at-risk or critical retention flag is present, unless a probe is explicitly requested. This conservative rule keeps a weakly supported score from suppressing assessment of a shared skill.

C. Performance and Mastery Update

For an item of difficulty $k \in \{1, \dots, 5\}$, the measurement value is $0.72 + 0.05k$ for a correct response, $0.12 + 0.035k$ for an unsure non-correct response, and $0.03 + 0.05k$ for any other incorrect one. The scoring API treats unsure as a non-correct response with no answer and rejects an answer submitted together with unsure. These values encode an ordinal

engineering judgement about difficulty and are not learned response probabilities. Each response then receives a weight

$$v_j = \text{clip}_{[0,1]}[(0.55 + 0.09k_j) \cdot (d_j/1.5) \ell_j g_j^* u_j]. \quad (4)$$

where d is clipped to $[0.25, 2.5]$, the cognitive weight ℓ is 0.62 for remember, 0.76 for understand, 0.92 for apply, and 1.00 for analyze, the guessing adjustment g^* is one minus the guessing prior for a correct response and one otherwise, and the certainty adjustment u is 0.72 for an unsure response and one otherwise. Skill performance P is the average of the measurement values weighted by $\max(0.1, v)$. Distinct item IDs are an operational independence check, not proof of statistical independence.

$$\rho = \min\{0.78, 0.16 \min(n, 3) + 0.06 \min(L, 3) + 0.08A + 0.06C\}. \quad (5)$$

Reliability ρ depends on the response count n , the number of distinct cognitive levels L , an indicator A for application or analysis evidence, and an agreement indicator C that is one when all responses are correct or all incorrect. Its ceiling of 0.78 deliberately stops a single session from being treated as conclusive, even if every answer is correct.

$$m' = (1 - \alpha) m_0 + \alpha P. \quad (6)$$

At the first diagnostic update, m_0 is the internal neutral prior and $\alpha = \rho$; when mastery already exists, $\alpha = 0.40\rho$. Updates are aggregated per skill at submission, so answer rows may share batch-level before-and-after values that are not independent per-question gains. Score improvements across sessions must not be reported as a treatment effect without a controlled study.

D. Confidence, Classification, and Uncertainty

Confidence combines evidence count, source diversity, session count, average difficulty, consistency, and the presence of retention evidence. The count term is $0.45(1 - \exp(-n/6))$; the other terms contribute up to 0.15 for source diversity, 0.12 for sessions, 0.06 for difficulty, 0.12 for consistency, and 0.10 for retention, on a base of 0.08. Consistency is one minus twice the population standard deviation of the stored per-item performance scores, clipped to $[0, 1]$, and the total is clipped to $[0, 1]$ and rounded to four decimals. If every evidence source is diagnostic, confidence cannot exceed 0.45, and with no evidence it is zero.

Fig. 2 summarizes the classification. No observation gives NOT_TESTED and one gives PROBED. Mixed correct and incorrect observations normally give NEEDS_CONFIRMATION. At least two negative observations with no correct response and mastery below 0.50 give GAP, and poor performance on a retention-sensitive check of previously strong knowledge can give FORGOTTEN. If an advanced skill scores 0.70 or higher while a required prerequisite from the same attempt is GAP or FORGOTTEN, the result is FRAGILE_FOUNDATION, so prerequisite

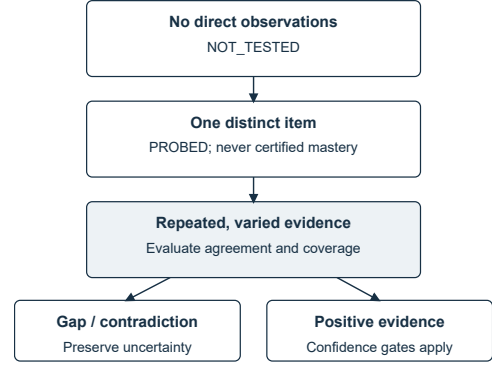


Fig. 2. Diagnostic classifications distinguish a probe from a supported decision. Coverage and evidence provenance are separate; positive evidence does not bypass confidence or prerequisite gates.

mastery is never fabricated. READY needs mastery of at least 0.65, confidence of at least 0.50, and two correct responses. MASTERED needs at least three observations, a correct application-level response, two cognitive levels, confidence of at least 0.70, and mastery meeting both 0.80 and the target. Under the current ceiling, diagnostic-only evidence therefore cannot produce a positive classification for a wholly new learner; the coverage planner has its own consistent-evidence route so assessment can progress without relabeling provisional knowledge as mastery.

$$B(m, c, n) = [\text{clip}(m - \delta), \text{clip}(m + \delta)], \quad (7)$$

$$\delta = 0.06 + \frac{0.22(1 - c)}{\sqrt{\max(1, n)}}.$$

The band in (7) is a heuristic visualization of uncertainty, not a calibrated 95% confidence interval or a Bayesian credible interval. It narrows by design as confidence or observation count grows, and its empirical coverage has not been established. Here clip bounds values to $[0, 1]$, c is clipped before use, and endpoints are rounded to four decimals.

E. Adaptive Stopping and Path Integration

A session normally needs at least 12 responses, fewer when the bank is smaller. The configured 16–22 question target is advisory and 28 is the hard maximum. Coverage requires $\min(K, \max(8, \lceil 0.75K \rceil))$ observed skills, where K is the active scope in the candidate set rather than the whole course, which means all four skills in a progressive session. Beyond coverage, the policy examines up to three highest-dependent prerequisite gateways. It continues while a gateway has an unresolved two-response disagreement, and it requires recognized evidence or at least two gateway observations (three if the evidence is mixed). A gateway with positive evidence also needs application or analysis evidence, and repeated negatives may preserve a lock. When these checks pass, the session stops as EVIDENCE_SUFFICIENT.

That label reflects an operational decision about coverage and selected gateways; it does not guarantee that non-

Algorithm 1 Eligibility-first research variant

- 1: Compute the same stage and scoring components as the deployed policy.
 - 2: Remove asked items and all items with non-finite scores.
 - 3: Apply coverage and contradiction preferences to this eligible set; restrict only when the preferred subset is nonempty.
 - 4: Apply role preference within each remaining skill, retaining its eligible fallback when the role is absent.
 - 5: Rank the remaining items with the unchanged score and tie-breaks.
 - 6: Return null only when the eligible set is empty.
-

gateway skills are resolved, and `MAXIMUM_REACHED` and `BANK_EXHAUSTED` stay distinguishable from it. After submission, the global Skill Passport shares new evidence with every enrolled course that uses the same skill. Required-prerequisite checks stay separate from diagnostic scope, so weak basics can lock advanced learning without blocking diagnostic probes of advanced skills. The knowledge-boundary summary picks the first course-ordered skill not supported by strong evidence, a conservative heuristic rather than a probabilistic frontier, and path regeneration then explains what changed the next action.

V. ELIGIBILITY-PRESERVING SELECTION

A. Preference Filtering Can Hide Eligible Questions

The deployed selector first drops questions already asked, then favors contradictory skills or a preferred diagnostic role, and only afterward removes candidates with non-finite scores caused by an observation cap. These steps do not commute. A preferred contradictory skill may already sit at its four-observation limit while another skill still has an eligible question; narrowing the pool first throws away the only usable alternatives, and the application service then logs `BANK_EXHAUSTED` although the physical bank is not empty.

The requirement is conditional availability: if at least one unasked item has a finite policy score in the current state, the selector must return an item. It need not ask every item or override the stopping policy; the rule applies only after the stopping policy has requested another question. As an illustration (not another experiment), take a verification stage in which s_1 has contradictory evidence but has reached its observation cap, while s_2 still has an unasked item with a finite score. Preference-first selection narrows to s_1 , removes its capped items, and returns null, hiding s_2 . Eligibility-first selection removes s_1 's capped items before applying preferences, so s_2 survives and is returned. The research variant changes a single candidate-pool expression so that it runs before the existing preferences; scoring weights, response processing, mastery updates, stopping rules, and confidence ceilings are unchanged. For a fixed state, let U be the set of unasked items with finite scores. Every later conditional preference keeps a nonempty subset whenever U is nonempty,

the role preference keeps at least one item per retained skill, and ranking a finite nonempty set always yields an item, so availability is preserved. This is a selection property, not a proof of optimal assessment or accurate estimation, and it says nothing about the feasibility of later selections under whole-test content or exposure constraints.

B. Scope of the Correction

The variant is an isolated source transformation applied by the research harness, not a production change, and it tests only whether eligibility ordering explains premature null selection. It does not fix shallow banks, a poorly calibrated learner model, or an inadequate stopping rule, and a session that stops as `EVIDENCE_SUFFICIENT` can still leave non-gateway skills uncertain.

VI. REPRODUCIBLE ENGINEERING EVALUATION

A. Implementation and Experimental Design

The deployed core is revision `b20a00e` with policy curriculum-evidence-adaptive-v6. A recheck on 24 September 2026 passed 182 API tests in 34 files and a read-only structural audit of 162 active diagnostic items across 42 skills; integration-test files are excluded, and these counts say nothing about learner validity. The replay runs the production pure functions for selection, stopping, estimation, confidence, and coverage without database access.

We generate 4,096 synthetic potential-response matrices from two bank depths, four response-propensity profiles, two noise settings, and 256 seeds per stratum. Each session scope has four skills, matching the production cap, and each bank holds three or six distinct items per skill. The six-item bank uses difficulty [1, 2, 3, 3, 4, 5], cognitive levels [remember, understand, apply, apply, analyze, apply], and roles [anchor, anchor, verification, verification, challenge, challenge]; the three-item bank keeps positions 1, 3, and 5. Discrimination is fixed at one, the guessing floor is 0.20, and the prerequisite dependent counts are [3, 2, 1, 0]. These banks are controlled stress-test settings, not a representative curriculum sample, and their metadata schedules exercise selector branches rather than reflect validated item calibrations.

The four profile vectors are [0.20, 0.20, 0.20, 0.20], [0.85, 0.85, 0.85, 0.85], [0.20, 0.85, 0.85, 0.85], and [0.85, 0.65, 0.35, 0.15], standing for low, high, weak-foundation, and uneven response propensity; they are not ground-truth mastery. For propensity θ and item difficulty k , the potential correctness probability is

$$p = 0.20 + (0.80 - \eta) \cdot \text{sigmoid}[\text{logit}(\theta) - 0.35(k - 3)]. \quad (8)$$

with noise η equal to 0 or 0.15. An independent unsure response with probability 0.10 suppresses correctness, mirroring the scoring contract, and nothing is learned or forgotten within a replay. Each item gets one fixed potential response per matrix that every method reuses, so this is a policy stress test, not a validated learner simulator. Skills with propensity of at least 0.60 report `COMFORTABLE` and the rest

TABLE I
SYNTHETIC POLICY COMPARISON

Policy	Questions	Assessed	Null*
Deployed adaptive	10.07	26.55%	47.02%
Eligibility-first	12.00	33.81%	0.00%
Stratified fixed-12	12.00	37.77%	0.00%
Stratified matched	10.07	30.46%	0.00%
No self-report	10.07	26.55%	47.02%
No graph components	10.08	27.90%	47.07%

*Premature null with an eligible alternative. Assessed is coverage, not accuracy. Each row uses 4,096 synthetic matrices.

NEVER_LEARNED, always with high self-report confidence. Mastery and confidence start unknown, which excludes returning learners. The protocol, seed rules, source hashes, generator, run-level CSV, aggregate JSON, and a replayable failure trace accompany the paper.

B. Baselines, Ablations, and Outcomes

Every matrix is run under six configurations. The deployed adaptive policy and the eligibility-first variant use the existing stopping rules with a maximum of 28 questions. Fixed-12 asks three stratified questions per skill in round-robin order, preferring anchor, application, and challenge items first. Matched asks as many questions as the deployed run asked on that matrix, in stratified order. No-self-report removes familiarity and confidence metadata, and no-graph sets dependent counts to zero in both selection and stopping, removing the graph-aware components together. All configurations share one estimator and coverage rule.

The main outcome is the fraction of the four skills labeled ASSESSED, which can include confirmed gaps. We also report question count, unresolved fraction, observed skills, positive READY/MASTERED declarations, and premature-null frequency, meaning the deployed selector returns no item while the eligibility-first selector finds an eligible one in the same state. A premature null is neither a false-mastery rate nor a human dropout rate. Paired mean differences use 2,000 stratified bootstrap replicates [15] that resample matrices within the 16 equal-sized strata, and the 95% percentile intervals describe simulation uncertainty only. No weights are fitted to these results. The analysis is exploratory and not externally preregistered; the local protocol records two harness corrections (enforcing the unsure contract and reusing the production confidence summarizer).

C. Aggregate Results and Non-Superiority

Table I reports all six configurations over 24,576 runs. The deployed selector asks 10.07 questions on average, but that is not efficiency: in 47.02% of runs it ends the session while an eligible alternative is hidden by preference filtering. In our data these failures occur only in the six-item banks (94.04% of those runs, against none with three items per skill). The eligibility-first ordering removes the failure, asking 12.00 questions on average and reaching 33.81% assessed-skill coverage against 26.55% for the deployed policy. That

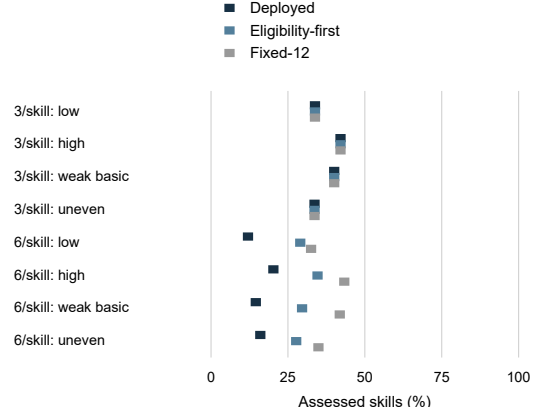


Fig. 3. Synthetic coverage by bank depth and response profile; the two noise levels are equally pooled. Each profile-depth group contains 512 matrices. No human accuracy is represented.

paired change of 7.26 percentage points has a synthetic 95% bootstrap interval of [6.87, 7.65].

Yet fixed stratified selection reaches 37.77% at twelve questions, so eligibility-first differs from it by -3.96 points, and the study does not support an adaptive coverage advantage. Fixing an availability defect and beating a reasonable baseline are different requirements. At question counts matched to the deployed policy, deployed minus stratified coverage is -3.91 points $[-4.33, -3.50]$. This holds burden closer to constant, but the matched budget is itself set by a failure-prone adaptive run, so it is not a randomized comparison of fixed budgets and says nothing about learning gains.

D. Sensitivity, Uncertainty, and Invariants

Removing self-report changes mean coverage by 0.00 percentage points [0.00, 0.00], and removing the graph-aware components changes it by 1.35 points [0.86, 1.88]. These are joint effects of the configured metadata, role filters, caps, and stopping policy, and a small or zero difference does not show that self-report or prerequisites are unimportant in real curricula. Because the cold-start design holds missing-mastery uncertainty at its maximum and gives every item identical priors, the information proxy is constant and the ablations cannot rank the coefficients; Fig. 3 and the per-stratum results keep the profile and depth differences visible.

All configurations observe all four skills, yet considerable skill-level uncertainty remains. No configuration emits a positive READY or MASTERED declaration, because diagnostic-only confidence cannot exceed 0.45 while the lowest positive threshold is 0.50. That zero is structural, not a measured false-positive rate; a learner study must measure whether knowledgeable learners can progress without excessive follow-up burden. A finite check enumerates 136 valid cases over one to four observations, four representative metadata schedules, binary correctness patterns, and all-unsure responses; none violates the confidence ceiling, the one-response PROBED rule, or the bounds of the uncertainty band, and none emits a posi-

tive cold-start classification. Planner traversal visits 100, 1,000, and 5,000 artificial skills in 25, 250, and 1,250 sessions within the four-skill scope, with selected scopes marked resolved by the harness; this shows first-pass scheduling reachability only, and we make no claim about latency, throughput, or course completion.

VII. DISCUSSION AND VALIDATION BOUNDARY

A. Interpretation

A shorter assessment can signal selection failure rather than sufficient evidence. The lesson is to separate physical bank exhaustion from candidate exhaustion caused by the policy: preference heuristics should operate inside an eligibility-preserving set so that a preferred but unusable skill cannot hide valid alternatives. This follows from the failure we demonstrated and is not a claim of universal optimality; availability, coverage, burden, estimation validity, and educational effectiveness are separate dimensions.

B. Threats to Validity

The simulator omits learning, fatigue, response time, correlated misconceptions, wording effects, and context-dependent transfer. The design space is narrow (four skills, two bank depths, fixed item priors, propensity-aligned self-report), the replay bypasses persistence, the interface, network failures, and concurrent learners, and finite invariant checks cannot replace integration tests, so the correction needs application-level verification before deployment. ASSESSED is an internal coverage state that includes confidently identified gaps, not an accuracy label. The confidence ceiling structurally rules out positive cold-start classifications and may trigger unnecessary reassessment. The uncertainty band is not a calibrated interval, and the bootstrap intervals describe this generator, not a learner population.

C. Independent Learner Evaluation

A follow-up study should recruit consenting learners and use expert-reviewed constructs and independent reference assessments that share no items with the system. It should prespecify an endpoint balancing incorrect readiness, unresolved placement, and question and time burden; compare corrected adaptive and stratified selection at comparable budgets; separate cold-start from returning learners; and estimate uncertainty at the learner level, with sample size from an endpoint-specific power analysis rather than the item-monitor gate of 20 responses. Retention and transfer claims need delayed checks and replication across subjects.

No participants were recruited and no private learner records were analyzed. Ethics approval, consent, and expert validation are not represented as completed. The local package provides code, source hashes, and synthetic outputs, and has no archival DOI. Human-effectiveness claims require a separate approved study.

VIII. CONCLUSION

LearnPath’s architecture makes the limits of its evidence inspectable, but that does not establish diagnostic validity. On RQ1, replay exposes a real preference-filtering defect, and eligibility-first ordering eliminates the premature null we observed. On RQ2, coverage improves over the deployed selector, yet fixed stratification stays higher at the same twelve-question budget, and the matched comparison also disfavors deployed adaptive coverage. On RQ3, the bank and profile results and the ablations describe behavior specific to this configuration, not the general value of any feature. Together, the architecture, the availability invariant with its correction, and a reproducible evaluation that keeps its negative results support an engineering fix, not adaptive superiority or diagnostic accuracy. Application-level integration and independent human validation remain future work.

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